

MAC21 — The 48-Hour Plan

Now: Sunday 12:00 · **Exam:** Tuesday 09:30 · **Real study time available:** ~22 hours · **Target:** 60–70 marks by the easiest route

The whole strategy in one line

In each unit, answer **a) + b) + ONE of the two 7-markers**. That is $2 + 4 + 7 = 13$ marks per unit $\times 5$ units = **65 marks**.

You do not need both 7-markers. That halves the hardest part of the syllabus and is why this is achievable in two days. Anything you manage on the second 7-marker is bonus on top of 65.

The 5 questions you are *likely* answering

Decide in the hall, not now. The question numbers below are where these topics have *usually* sat — they are a tiebreaker, not a commitment. Topics do not vanish when a paper is reshuffled, they just move. If "expand about $(1, \pi/2)$ " turns up in Q1 instead of Q2, you answer Q1. **Study the topics; choose the question after reading both options.**

UNIT	ANSWER	A) — 2 MK	B) — 4 MK	YOUR ONE 7-MARKER	TOTAL
I	Q2	State Maclaurin's / Taylor's series	$xy(1-x-y)$ max at $(1/3, 1/3)$	Expand $xy+\cos(xy)$ about $(1, \pi/2)$ 4x	13
II	Q4	DE of L-C circuit / define orthogonal trajectories	Bungee Euler <i>or</i> Taylor's method <i>or</i> orthogonal trajectory	Modified Euler's method 5x (<i>sits at 4c in 4 of 5</i>)	13
III	Q5	Steps for solving Cauchy's LDE	Prove $1/(D+a) \cdot X$ 4x	Cauchy's equation 4x (<i>easier than variation of parameters</i>)	13
IV	Q8	Lagrange's inverse interpolation formula	Find the missing terms 3x	Newton's divided difference 5x <i>or</i> Simpson's 3/8	13
V	Q10	Rank / echelon / consistency definition	Gauss-Seidel, 2 iterations 3x	Gauss elimination word problem (<i>easier than system of ODEs</i>)	13
Floor if every 7-marker lands					65

What you are deliberately NOT studying. Every one of these lives in the question you are *not* answering. Touch them only if you finish everything else:
Legendre's equation (Q6) · **variation of parameters** (the other 7 in Q5 — learn only if time) · **Rayleigh's power method** (Q9) · **diagonalization / modal matrix** (Q9) · **Lagrange multipliers** (Q1) · **maxima-minima beyond $xy(1-x-y)$** (Q1) · **radius of curvature, Newton-Cotes derivation, backward difference tables**.
That is roughly 40% of the syllabus, cut on purpose.

How exposed are you really? — syllabus coverage audit

This checks the plan against **your 28 sessions**, not against the predictions. It is the honest answer to "what if the paper ignores the prediction sheet".

UNIT	SESSIONS	PLAN ALONE	+ INSURANCE	WHERE THE EXPOSURE IS
II	5	5 / 5	5 / 5	Zero exposure — and this is arithmetic, not a prediction. Unit II has 5 sessions and four of them <i>are</i> the numerical methods. No question can be set that avoids them.
I	6	4 / 6	6 / 6	Missing Lagrange multipliers and 1-variable expansion. Both cheap to add.
III	6	4 / 6	6 / 6	Cauchy <i>forces</i> you to learn CF + P.I. anyway, so sessions 12–14 come free. Missing Legendre and variation of parameters — and Legendre is 20 minutes once Cauchy is in.

IV	6	4 / 6	6 / 6	Missing numerical differentiation and Newton-Gregory as a full problem. Both are formula-application.
V	5	2 / 5	5 / 5	This is the real hole. No eigenvalues, so no diagonalization, no Rayleigh, no system of ODEs. If both Q9 and Q10 lean eigen-heavy you are short.
Total		~19 / 28	28 / 28	Plan alone $\approx$ 68% of the syllabus. With the insurance block below, effectively all of it.

The 90 minutes that removes most of the risk

Eigenvalues and eigenvectors. Sessions 26, 27 and 28 are *one chain* — diagonalization, Rayleigh's power method and system-of-ODEs-by-matrix are all "find the eigenvalues, then do one more step". Learn the eigen-step once and all three unlock together.

One 90-minute block takes Unit V from 2/5 to 5/5. It is the highest-leverage block in the whole two days.

SUNDAY — 15:00 to 00:00 (5.5 h study)

Only two blocks today, so they go to the **two most predictable units**: III and V. Both are heavily locked and both reward a fresh brain.

TIME	BLOCK	WHAT EXACTLY
15:00–18:00	UNIT III 13 mk	The biggest single win in the paper. Do it while you are freshest. • CF + P.I. for constant coefficients first — everything else in this unit sits on it. • Memorise the $1/(D+a)$ proof cold. 4 marks, appeared 4x, never changes. • Cauchy's equation — substitution $x = e^z$, then it is just the constant-coefficient problem you did an hour ago. Work M23 5c, M24 5d, MAM24 5d. • Write the "steps for solving Cauchy's LDE" answer out once.
18:00–21:30	Break	Long one. Eat properly. Do not half-study through it — you have a late block coming.
21:30–23:30	UNIT V core 13 mk	Mechanical work, which suits a late slot. • Gauss-Seidel — start with the MP/P26 system ($x+8y+2z=-4 \dots$), it appeared twice identically. Then M23 9b and M24 10b. Three reps and it is automatic. • Gauss elimination — M23's electrical network, M24's dietician problem. • Rank, echelon form, consistency conditions.
23:30–00:00	Consolidate	Notes closed. Write from memory: the $1/(D+a)$ proof, and the Unit III + Unit V definitions. Anything that will not come out, star it — it is your first job tomorrow.
00:00–07:00	Sleep	7 hours. Hold this line.

End of Sunday you own Units III and V core — 26 marks.

MONDAY — 10:30 to 23:00 (~10.5 h study)

The heavy day. A workout in front of it is fine — it helps retention — but **eat before you start at 10:30**, and treat the 16:00 and 18:00 breaks as compulsory. This is a 12-hour day and it only works with them.

TIME	BLOCK	WHAT EXACTLY
07:00–10:30	Wake, workout, eat	Your call — but arrive at the desk fed.
10:30–13:00	UNIT IV 13 mk	• Missing terms — do the $45\dots65 / 3.0, -, 2.0, -, -2.4$ table; it appeared identically twice. • Newton's divided difference — the $(2,4)(4,56)(9,711)(10,980)$ cubic appeared twice verbatim. Do that exact one. • Simpson's 1/3 and 3/8 on $\int_0^1 dx/(1+x^2)$ — your backup 7-marker. • Lagrange's inverse interpolation formula. Write it out.
13:00–14:00	Lunch	—

14:00–16:00	UNIT II 13 mk	Learn the four methods <i>as one family</i> — they share a skeleton, so together is far faster than one at a time. • Taylor's method → Euler's → Modified Euler's → RK4. Tabulate, iterate, 4 decimals. • Modified Euler is your 7-marker (slot 4c, 4 of 5 papers). Do three. • Newton's law of cooling — 10 minutes, appeared 5x . Do the coffee problem. • Orthogonal trajectories, Cartesian and polar. The circuit-DE 2-marker.
16:00–16:20	Break	Compulsory.
16:20–18:00	UNIT I 13 mk	• Expand about $(1, \pi/2)$ — appeared 4x , the point never changes. Do all three variants. • $xy(1-x-y)$ maximum at $(1/3, 1/3)$ — 3x . Learn the r-s-t test on it. • The N-R system $x^2+y^2=x$, $x^2-y^2=y$ from $(0.8, 0.4)$ — 4x . Your escape hatch if the other option looks bad. • Maclaurin one variable, Taylor two variables — the statements.
18:00–19:00	Dinner	Compulsory.
19:00–21:00	INSURANCE do not skip	This is what makes the plan survive a reshuffled paper. In this order: • Eigenvalues & eigenvectors — 90 min. Unlocks diagonalization, Rayleigh's power method <i>and</i> system of ODEs by matrix method together. Unit V goes 2/5 → 5/5. Highest-value block of the weekend. • Legendre — 20 min. Same substitution as Cauchy with $(ax+b) = e^z$. Nearly free now, and it closes Q6. • IVP / BVP — 10 min. Free once you have CF + P.I.
21:00–22:15	MOCK, shortened critical	Model paper, timed, closed book, 75 minutes : all five a) and b) parts, plus your one 7-marker in Units III and V. Shortened because you no longer have time to both sit a full mock <i>and</i> fix what it finds — and finding is only useful if you can fix.
22:15–23:00	Repair + formula sheet	Redo whatever you dropped in the mock — the actual question, not "review the method". Then write every formula you need on one sheet, by hand, from memory.
23:00–06:00	Sleep	7 hours.

TUESDAY — 06:00 to 09:30

TIME	BLOCK	WHAT EXACTLY
06:00–06:40	Wake, breakfast	Eat properly. Do not skip it.
06:40–07:40	2-mark bank, all five units	Every definition, from memory, written out. 10 marks and it should be the only thing still wobbling.
07:40–08:30	Setups only	For each 7-marker write just the <i>first three lines</i> , not the full solve: $1/(D+a)$ proof · Cauchy substitution · Modified Euler table · divided-difference table · Gauss-Seidel rearrangement. Ten minutes each.
08:30–08:50	Skim	Tier 1 table of the prediction sheet. Read, do not solve.
08:50	STOP	Close everything. Travel. Nothing new after this point — it displaces what is already in place and costs more than it adds.

What the shorter schedule costs you. Against the 22-hour version you lose roughly 4 hours, and these come out:

- **Variation of parameters** — acceptable. It is the *other* 7-marker in Q5 and you only need one; Cauchy is the easier of the two. This was already first in the sacrifice order.
- **Lagrange multipliers, 1-variable expansion** (Unit I) and **numerical differentiation, Newton-Gregory problems** (Unit IV) — leaves those two units at 4/6 rather than 6/6.
- The full 2-hour mock becomes 75 minutes.

Coverage lands near **24 of 28 sessions** instead of 28. The 65-mark floor is intact; your buffer against a strange paper is thinner. **If you find spare time anywhere, spend it on Lagrange multipliers** — it appeared **4x** and is the largest remaining gap.

In the exam hall — 09:30 to 12:30

MINUTES	DO THIS
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0–5	Read all ten questions before writing anything. In each unit compare the two options and commit. Write your five choices on the top of the answer sheet so you do not drift.
5–20	All five 2-markers first. 10 marks in 15 minutes, while you are calm. Nothing else in the paper has that rate of return.
20–50	All five 4-markers. Another 20 marks. You are now at 30 with 100 minutes left.
50–170	The 7-markers — easiest first. 24 minutes each $\times 5 = 120$. Do your strongest unit first to build momentum. If one stalls past 25 minutes, leave it and move on; come back only if there is time.
170–180	Sweep. Check every question is numbered and lettered correctly. Attempt the second 7-marker anywhere you have a spare minute — partial marks are still marks.

Show every step. These are method-marked papers. A Gauss-Seidel table with an arithmetic slip in the last row still scores most of the 4. A correct final answer with no working does not. **Never leave a part blank** — write the formula and the setup even if you cannot finish; that is 2–3 marks on a 7-marker for ninety seconds of work.

Sleep 7 hours both nights. This is the highest-leverage instruction on the page. Studying until 2am costs you more in recall and arithmetic accuracy than the extra hours add — and this paper is almost entirely arithmetic accuracy. If you are behind on Monday night, cut the bonus 7-markers, not the sleep.

If you open the paper and it looks unfamiliar anyway. Do not panic-read all ten questions twice — work the structure instead.

- The five 2-markers are drawn from a small fixed pool of definitions. Take them first regardless of how the rest looks. That is **10 marks** banked before you have solved anything.
- In each unit, count how many parts of *Qodd* vs *Qeven* you can start. Pick on the count, not on which looks nicer.
- These are method-marked. On any 7-marker you cannot finish, write the formula, the setup, and the first two steps — that is routinely 3 of 7 for ninety seconds' work. **Five partial attempts beat two perfect answers and three blanks.**
- A "new" question is usually a familiar method in unfamiliar clothing. Radius of curvature was an interpolation problem. A dietician's meal plan was Gauss elimination. Ask "what data am I given?" before "what topic is this?"

If you fall behind — the order to sacrifice things:

- Variation of parameters (last item of the insurance block) — drop it, the first three items matter far more.
- Unit I depth — keep only "expand about $(1, \pi/2)$ " and $xy(1-x-y)$.
- Simpson's 3/8 backup in Unit IV — divided difference alone is enough.

Never sacrifice: the 2-mark bank · the $1/(D+a)$ proof · Gauss-Seidel · missing terms · Modified Euler · **and the eigenvalue block.** Those are ~25 marks plus your entire Unit V safety net, for about five and a half hours of work.

Built from the MAC21 prediction sheet: 6 papers analysed, repeat counts stated out of the papers carrying each unit. Question-choice recommendations reflect observed slot frequency, not certainty — always read both options before committing. Timings assume you start at Sunday 12:00 and are asleep by 23:00 both nights.